

## Simple Belief Elicitation: Proofs

### Proposition 1

*Proof* To prove the “only if” statement suppose  $b$  maximises  $f(b)$ . If  $p_v = 0$  then it is clearly best if  $b_v = 0$  because if  $b_v > 0$  the prize will be won with probability 0. For any  $u \neq v$  with  $b_v, p_v > 0$ ,

$$\begin{aligned} f(b_1, \dots, b_u, \dots, b_v, \dots, b_k) - f(b_1, \dots, b_u + 1, \dots, b_v - 1, \dots, b_k) &\geq 0 \\ \Rightarrow \frac{n!}{b_1! \dots b_k!} \prod p_i^{b_i} - \frac{b_v p_u}{(b_u + 1) p_v} \frac{n!}{b_1! \dots b_k!} \prod p_i^{b_i} &\geq 0 \\ \Rightarrow f(b) \left( 1 - \frac{b_v p_u}{(b_u + 1) p_v} \right) &\geq 0 \end{aligned}$$

which gives us the set of constraints

$$b_v p_u \leq (b_u + 1) p_v \forall u \neq v. \quad (1)$$

Now  $p_i = \sum_j \frac{b_j}{n} p_i = \frac{b_i}{n} p_i + \sum_{j \neq i} \frac{b_j}{n} p_i \leq \frac{b_i}{n} p_i + \sum_{j \neq i} \frac{b_i + 1}{n} p_j = \frac{b_i}{n} + \frac{1}{n} (1 - p_i)$  which implies

$$p_i \leq \frac{b_i + 1}{n + 1}. \quad (2)$$

Also, for  $b_i > 0$ ,  $p_i = 1 - \sum_{j \neq i} p_j \geq 1 - \sum_{j \neq i} \frac{(b_j + 1) p_i}{b_i} = 1 - \frac{p_i}{b_i} (n - b_i + k - 1)$ , which implies

$$p_i \geq \frac{b_i}{n + k - 1}. \quad (3)$$

To prove the “if” statement assume that  $b$  satisfies the constraints given in the proposition. Consider any  $b'$  such that  $f(b') > 0$ ,  $b'_u > b_u$  and  $b'_v < b_v$ . Hence,  $p_v > 0$ . From the above equations above we obtain

$$\begin{aligned} &f(b'_1, \dots, b'_u, \dots, b'_v, \dots, b'_k) - f(b'_1, \dots, b'_u + 1, \dots, b'_v - 1, \dots, b'_k) \\ &= f(b') \left( 1 - \frac{b'_v p_u}{(b'_u + 1) p_v} \right) \\ &> f(b') \left( 1 - \frac{b_v p_u}{(b_u + 1) p_v} \right) > 0. \end{aligned}$$

This means that whenever we increase the report of event  $u$  by one and at the same time decrease the report of  $v$  by one then the probability of winning the prize goes down, provided the report of  $u$  was above  $b_u$  and the report of  $v$  was below  $b_v$ . Thus, for any given  $p$  we can compare  $f(b)$  to any other  $f(b')$ , by repeating the above for all  $u \in \{i : b'_i > b_i\}$  and  $v \in \{i : b'_i < b_i\}$ . This shows that  $b$  maximizes  $f$  over all  $b' \in B$  which completes the proof.

### Proposition 3

*Proof* Note that

$$f(m) = \binom{n}{n/2} (P(Y \leq m) \cdot (1 - P(Y \leq m)))^{n/2} \quad (4)$$

Since  $x(1-x)$  is maximized if and only if  $x = 1/2$  we obtain that  $f$  is maximized if and only if  $P(Y \leq m) = 1/2$  which completes the proof.

### Proposition 4

*Proof* Following the proof of Proposition 3,  $f$  is maximized by  $x_m$  if and only if  $|P(Y \leq x_m) - \frac{1}{2}| \leq |P(Y \leq x) - \frac{1}{2}|$  for all  $x \in X$ . Let  $x_i$  be the median of  $Y$ . If  $P(Y \leq x_i) = 1/2$  then  $x_m = x_i$ . If  $P(Y \leq x_i) > 1/2$  then  $x_m \in \{x_{i-1}, x_i\}$  as  $P(Y \leq x_{i-1}) < 1/2$ .

It is easy to see that the proof also goes through if one allows for multiple medians by defining  $m_{1/2}$  to be a median of  $Y$  if  $P(Y \geq m_{1/2}) \geq 1/2$  and  $P(Y \leq m_{1/2}) \geq 1/2$ . One only has to adjust the statement of the proposition, replacing “the median” by “a median”.